purple8

PURPLE8, INC. · 2026

AI-Native Execution Infrastructure.

Built for AI, Operated by AI and Supervised by Humans.

Infrastructure that enables AI systems to autonomously build, execute, and operate enterprise applications with native intelligence, adaptive computing, and human-supervised governance.

Read · Remember · Reason · Answer · Run · Protect · Prove · One Engine built to solve complex problems.

1 Engine · 0 External DB Dependencies

120+ MCP Tools · 16 Namespaces

Proven at 20M Nodes & 30M Edges on 1 Node

Nikhil Dutt, Founder & CEO · hello@purple8.ai · www.purple8.ai

THE PROBLEM

AI models can reason, but they have no native infrastructure to execute.

THE 2026 "STITCH-AND-PRAY" AI STACK

Pinecone / Weaviate (Vectors)	Separate cluster + bill
Neo4j (Knowledge Graph)	Separate JVM cluster
PostgreSQL / MongoDB (State)	Separate DB + DBA ops
Airflow / Temporal (Workflows)	Separate orchestrator
LangChain + LangGraph + LangSmith	Endless glue code
Unstructured / Cloud OCR (Docs)	Per-page cloud tax
Auth0 + KMS + Elasticsearch + SIEM	4 more services

1. Built for human microservices, not AI agents

Shipping one production AI app forces teams to wire together 8 to 12 separate services. AI agents cannot operate them directly without brittle human-written glue code.

2. Unbounded memory & runaway cloud bills

Traditional vector and graph databases scale RAM linearly with corpus size, turning every data doubling into a costly cluster re-provisioning event.

3. Broken governance across 10 silos

When context, documents, and workflow states live in separate databases, enforcing real-time SLAs, Human-In-The-Loop approvals, and tamper-proof audit trails is nearly impossible.

THE SOLUTION · ONE ENGINE, SEVEN CAPABILITIES

One unified engine that reads, remembers, reasons, answers, runs, protects, and proves.

PURPLE8 CORE · FLUID STATE ARCHITECTURE

0 Network Hops · In-Process

1. READ · DocIntel
70+ doc formats & 75+ languages into graph entities

2. REMEMBER · Cortex Storage
Graph + vector + document + full-text + temporal in one engine

3. REASON · Algorithm Library
Ranking, communities, paths, resilience & multi-objective optimisation

4. ANSWER · Self-Tuning RAG
Hybrid GraphRAG with groundedness & source citations

5. RUN · Journey Engine
Stateful workflows, SLA monitors, HITL gates & scheduling

6. PROTECT & 7. PROVE
AES-256 KMS, 4-tier RBAC, threat containment & immutable audit

Fluid State Architecture: Evolve schemas & workflows live with zero migrations

`docker run purple8/hyper-graph`

ARCHITECTURAL IMPACT

DIMENSION	LEGACY AI STACK	PURPLE8
Services per app	8 to 12 processes	**1 process**
Network hops / query	6 to 14 hops	**0 (in-process)**
HA server footprint	24 to 36+ nodes	**3 nodes**
Memory scaling	Unbounded O(N)	**Hardware-bounded**
Agent interface	Custom glue code	**120+ MCP tools**
Audit & HITL gates	Bolted on later	**Native graph edges**

WHY NOW · THE PARADIGM SHIFT

The AI agent is the developer.

Agent-Built Applications on Governed Infrastructure.

120+ MCP TOOLS ACROSS 16 NAMESPACES

Because Purple8 is MCP-native from day one, any AI agent (Claude, Cursor, Copilot, Codex) can build schemas, ingest documents, run hybrid RAG, enforce SLAs, and route human approvals without a human writing backend glue code.

- ✓ **Focused MCP Tool Profiles:** Connect with `?profile=starter|builder|ops|full` so agents get the exact toolbelt they need.
- ✓ **Zero-Install MCP Connection:** Point Claude Desktop, Cursor, or VS Code at your Purple8 Docker container or Cloud Sandbox URL and start building in seconds.
- ✓ **Human-Supervised Governance:** Destructive and credential-modifying tools enforce `human_only=True` and storage-layer HITL gates.

```
MCP Endpoint: https://<your-purple8-instance>/mcp?profile=starter
```

1. Domain Expert or Developer States Intent
"Define a loan origination workflow with a 24h KYC SLA and underwriting HITL gate, then ingest this borrower packet."

↓ MCP Streamable HTTP / SSE

2. AI Agent Calls Purple8 MCP Primitives
`schema.define_node_type` → `journey.define` → `data.ingest_file` → `rag.hybrid_query`

↓ Zero Backend Glue Code Written

3. Production Backend Live Immediately
Encrypted, multi-tenant, SLA-monitored, and hash-chain audited inside a single Purple8 container.

THE "SO WHAT" · FUNCTIONAL EQUIVALENCE WITHOUT THE COMPLEXITY TAX

Anyone can stitch this together functionally.
Only Purple8 delivers it without the cost & complexity.

1. Hardware-Topology & Accelerator Adaptive

B200/B300 · NUMA · AMX

Automatically senses and adapts to CPU sockets, NUMA domains, Intel AMX/XPU, Apple Unified Memory, and NVIDIA Hopper/Blackwell GPUs (B200/B300 FP4 up to 288 GB VRAM), maximising throughput per watt with zero manual tuning.

2. Bounded-Memory Storage Architecture

Constant RAM Envelope

Peak memory is governed by available hardware rather than dataset size N. Legacy vector and graph DBs scale RAM linearly with data; Purple8 runs 2M or 20M+ nodes within a fixed hardware envelope with zero cluster re-architecture.

3. Zero-Hop In-Process Reasoning Loop

0.87 ms P99 2-Hop

Multi-hop risk and network simulation require thousands of tight iterations across graph topology, vectors, and flow math. Across 10 networked services, latency compounds into seconds; in-process inside Purple8, it completes in milliseconds.

4. Self-Tuning Sentinel & Decision Gate

89% Fewer LLM Calls

In-engine multi-objective optimisation adapts thresholds to live outcomes and triages routine events deterministically before calling an LLM, cutting inference cost and energy by ~89% with zero missed high-risk cases in AML backtests.

THE "SO WHAT": Anyone can theoretically replicate what Purple8 does functionally, if they are willing to stitch together 10 vendors, hire a platform team, and pay a permanent tax in cloud compute, latency, and brittle glue code. Purple8 collapses that entire cost and complexity into one engine that an AI agent can operate on day one.

VERIFIED PERFORMANCE · SINGLE-NODE PHYSICAL CEILING

Proven at 20M nodes & 30M edges on a single cloud instance.

50M
GRAPH OBJECTS ON 1 NODE
20M nodes + 30M edges · 25.35 GB NVMe

52 µs
POINT LOOKUP P50
P99: 1.26 ms · 2-hop P99: 0.87 ms @ 20M

40,551/s
BATCH INGESTION
916 QPS @ 0.946 recall on 1M×768d vectors

0% Err
SOAK & CHAOS TESTED
1.18M requests · 500 concurrent users

LIVE SINGLE-NODE BENCHMARK RESULTS (16-VCPU CLOUD INSTANCE · OCTOBER 2026)

WORKLOAD TRACK	DATASET SCALE	SUSTAINED THROUGHPUT	QUERY / ANALYTICAL LATENCY
Mega-Scale Structured Graph	20,000,000 nodes · 30,000,000 edges	**3,453 nodes/s · 9,291 edges/s (<1.1% drift)**	**Lookup P50: 52 µs (P99: 1.26 ms) · 2-hop P50: 0.38 ms (P99: 0.87 ms) · 4,180 QPS (P99: 13.8 ms)**
Power-Law Super-Hub Traversal	6,000,000-edge central hub @ 20M scale	**100% completion · 0 timeouts**	**P50: 53.1 ms · P95: 58.8 ms · P99: 58.8 ms guarded 2-hop walk**
Unstructured Docs + Tables (DocIntel)	70+ formats · 75+ languages · FK linking	**353.4 pages/s · 70.7 tables/s (8w)**	**Table row query P50: 0.65 ms (P99: 1.12 ms) · Indexed scan P50: 39 µs (P99: 84 µs)**
3-D Structural Engineering (AEC)	250-storey frame · 100,000 3-D joints · 478K members	**352.5 MB peak memory**	**240 ms 3-D rigidity · 3.53 s critical-member detection**

COMMERCIAL WEDGE · START WHERE FAILURE IS EXPENSIVE

Financial Crime, Lending & Risk Intelligence running in a single engine.

WHY PURPLE8 IS PURPOSE-BUILT FOR BFSI

- ✓ **Unstructured KYC & Loan Ingestion:** DocIntel parses bank statements, KYC packets, corporate filings, and financial tables into connected graph entities.
- ✓ **Multi-Hop Fraud Ring & Laundering Detection:** Unmasks 5+ layer shell-company beneficial ownership, circular fund flows, and shared-attribute mule rings in <100 ms.
- ✓ **89% Alert Noise & LLM Cost Reduction:** Deterministic Decision Gate + Sentinel evolutionary threshold tuning cut false-positive triage fatigue dramatically.
- ✓ **Regulator-Ready Workflows & Audit:** Pre-built `loan_application` and `compliance_review` Journeys with SLA breach alerts, HITL sign-off, and hash-chained audit logs.
- ✓ **Zero Rip-and-Replace Pilot:** Deploys as an air-gapped Docker container alongside existing core banking/LOS systems; ingests KYC + transaction extracts in <48 hours.

< 100 ms
5-Hop Beneficial Ownership & Mule-Ring Unmasking
Across millions of accounts, devices, UBO filings, and transactions.

~89%
Reduction in Downstream AI Model Calls
Via in-graph deterministic risk gating (0 missed high-risk cases in AML backtests).

100% Air-Gap Capable
$0 Data Egress · Built for Regulated Finance
Runs inside the institution's VPC or on-premise hardware with customer-managed KMS.

THE COMPOUNDING HORIZON · LEAST COMPLEXITY, FASTEST TIME-TO-MARKET, SUSTAINABLE BY DESIGN

Empowering AI agents to build complex systems that solve humanity's hardest problems.

1. Least Complexity for AI Agents: Agents call atomic MCP primitives instead of writing 50,000 lines of distributed glue code.

2. Least Time to Market: Complex, governed systems go from idea to production in hours or days instead of 12-month SI projects.

3. Sustainable Economics & Energy: Bounded RAM + 89% fewer LLM calls make real-world deployments viable everywhere.

🌾 Food Security & Supply Chains

Agents model farm-to-distribution networks, detect chokepoints before shortages cascade, and dynamically reroute perishable supply across regions.

💧 Water Networks & Fault Isolation

Built-in max-flow/min-cut and fault-isolation traversal let agents pinpoint pipe failures, contain contamination blast radiuses, and route reserves in milliseconds.

⚡ Energy Grids & Load Balancing

Combinatorial and evolutionary solvers balance distributed renewable generation, storage dispatch, and feeder routing across cost and reliability objectives.

🏗️ Resilient Shelter & Infrastructure

12 native AEC & spatial modules let agents verify 3-D structural rigidity, minimise embodied material via topology optimisation, and simulate thermal energy balance.

🌍 Climate & Resource Resilience

Consolidating 10 always-on cloud clusters into one bounded-memory engine slashes AI's own carbon and DRAM footprint while powering regional adaptation twins.

One Engine. Two Products.

Purple8 Hyper Graph + **Purple8 DocIntel** remove the infrastructure barrier so AI agents can solve what matters most.

AI GOVERNANCE FOR LLMS & AGENTS · BUILT IN, NOT BOLTED ON

Most AI governance is an afterthought.
In Purple8, the execution engine is the governor.

14,500+
AUTOMATED TESTS
Unit, chaos, soak & security

125
GOVERNED MCP TOOLS
Per-tool RBAC & human_only gates

4-Tier
AGENT CONTAINMENT
In-process SOC behavioural profiler

100%
DOGFOODED LIVE
cc.purple8.ai + Cloud Sandboxes

WHY BOLT-ON GOVERNANCE FAILS

- ✕ **Post-hoc wrappers:** External proxy tools only log prompts *after* an agent has already mutated a database or leaked PII.
- ✕ **Blind to data & state:** An external proxy cannot see graph lineage, record classifications, or workflow SLA/approval state.
- ✕ **Fragmented audit logs:** Stitching traces across 10 vendors fails regulatory scrutiny under EU AI Act, DORA, and SR 11-7.

PURPLE8 NATIVE IN-ENGINE GOVERNANCE

- ✓ **Pre-Execution Agent Gates:** Per-tool RBAC, data-classification scopes, `human_only=True` blocks, and storage-layer HITL approvals.
- ✓ **LLM Policy Router, Guardrails & Drift Sentinel:** Routes LLMs by PII/stakes, redacts sensitive data, and flags model behavioural drift automatically.
- ✓ **Real-Time SOC & Hash-Chained Audit:** Autonomous containment of rogue agents + tamper-evident cryptographic audit edges for every AI and human action.

MARKET & BUSINESS MODEL

Container-first deployment. Enterprise-locked expansion.

$87B
AI INFRASTRUCTURE TAM
38% CAGR through 2030

$16.2B
GRAPH + VECTOR + DOCINTEL SAM
Immediate replacement budget

LICENSE-BASED MODEL · LAND, EXPAND & RETAIN

→ **1. Land (Docker Container & Cloud Sandboxes):** Instant evaluation via self-serve cloud sandboxes or a single hardened `docker run` container image with zero-edit MCP setup.

→ **2. Expand (Red Hat, AWS, Azure & GCP):** Capacity-tiered commercial and enterprise licenses procured directly or through **Red Hat**, **AWS**, **Azure**, and **GCP** Marketplaces.

→ **3. Retain (Compounding Gravity):** Every workflow, schema, and audit record lives inside Purple8, making it the permanent system of intelligence.

WHY CONSOLIDATION WINS ON UNIT ECONOMICS & ESG

Running 10 separate databases and orchestrators 24/7 with 3x HA replicas for every AI app is economically and energetically unsustainable.

METRIC	STITCHED STACK	PURPLE8
Always-on processes	8 to 12	**1**
RAM at 20M nodes (50M objects)	128 to 512 GB+	**~42.3 GB (25.35 GB NVMe)**
LLM triage calls	100% of events	**~11% (Decision Gate)**
Infra TCO savings	Baseline	**60% to 80% lower**

COMPETITIVE LANDSCAPE

Point solutions solve slices.
Purple8 owns the execution layer.

CAPABILITY	PINECONE / QDRANT	NEO4J	LANGGRAPH / TEMPORAL	PALANTIR FOUNDRY	PURPLE8
Native Property, Hyper & Temporal Graph	No	**Property only**	No	**Yes**	**Yes (In-process, ~7 µs writes)**
Semantic Vector + Hybrid GraphRAG	**Vectors only**	Bolt-on	No	Bolt-on	**Topology-Guided + Auto-Tune**
Bounded-Memory Envelope	No (grows with N)	No (JVM heap)	N/A	No (heavy cluster)	**Yes (Hardware-bounded Cortex)**
Stateful Workflows + SLA + HITL	No	No	**Yes (code-heavy)**	**Yes**	**Built-in Journey Engine**
70+ Format Document Intelligence	No	No	No	**Yes**	**Built-in (Purple8 DocIntel)**
Agent-Operated (120+ MCP Tools)	Read-mostly	Query-only	No	Proprietary UI	**Full Backend Built via MCP**
Deployment & Time to Value	Cloud SaaS slice	Cluster ops	Weeks of coding	Multi-month SI project	**1 install · Live in 60 seconds**

FOUNDER-MARKET FIT & THE ASK

Built by an operator who lived the problem. Raising $3M Seed to scale enterprise GTM.

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Nikhil Dutt
Founder & CEO · Purple8, Inc. · linkedin.com/in/nikhildutt

22+ years leading enterprise AI, ML, and cloud infrastructure programmes at HSBC across global risk operations, enterprise intelligence, and large-scale regulated delivery. Lived first-hand the cost of fragmented vendor stacks, compliance bottlenecks, and brittle 3 AM production incidents.

Built Purple8 from first principles to collapse that stack into one engine, with prior domain work spanning AI-native AEC platforms and **GeoHarvestAI** (sovereign agricultural intelligence at state-government scale).

Unmatched Execution Velocity
14,500+ automated tests, 20M-node single-instance benchmarks, self-serve cloud sandboxes, and enterprise Docker distribution shipped in under 12 months.

nikhildutt@purple8.ai · www.purple8.ai · linkedin.com/in/nikhildutt

$3M SEED ROUND · 18-MONTH MILESTONES

- ✓ **Enterprise GTM (40%):** Close first 15 enterprise license customers across BFSI (AML, KYC, lending workflows), Insurance, and AEC/BIM.
- ✓ **Core Systems Engineering (35%):** Multi-node HA clustering, federated cross-tenant supergraphs, and B200/B300 + NUMA hardware acceleration.
- ✓ **Ecosystem & Marketplaces (15%):** Scale container distribution and enterprise fulfilment across Red Hat, AWS, Azure, and GCP.
- ✓ **Security & Certifications (10%):** Complete SOC 2 Type II and ISO 27001 certification audits on top of our existing cryptographic controls.

Target Milestone: 15+ Enterprise License Deployments & Series A Readiness
Built for AI, Operated by AI and Supervised by Humans.